Multivariate Calculus Lecture Notes
Section 15.2 – Double Integrals over More General Regions

I. What if the base of the solid isn't rectangular?
   a. For example, what if we were trying to find $\iint_R f(x,y)dA$, where $f(x,y) = x^2y^3$, and R looks like this:

      i. Integrating with respect to x first
         - Fix a particular value of y.
         - For this fixed value of y, x ranges from the line $y = x$ to the line $x = 1$.
         - So the x bounds are from y to 1.
         - After we've integrated with respect to x, x is gone, and so we can collapse the region onto the y-axis.
         - Thus our y bounds are from 0 to 1.
         - And so we get $\int_0^1 \int_y^1 x^2y^3 dx\, dy$.
      ii. If we integrate with respect to y first we follow the same thought process to obtain $\int_0^1 \int_0^x x^2y^3 dy\, dx$.
      iii. Both iterated integrals will give us the same answer.
      iv. The second one looks a little easier, as it has more 0's in it.
      v. We have $\int_0^1 \int_0^x x^2y^3 dy\, dx = \int_0^1 \frac{1}{4}y^4\Big|_0^x x^2 dx = \frac{1}{4}\int_0^1 (x^4)x^2 dx = \frac{1}{4}\int_0^1 x^6 dx = \frac{1}{28}x^7\Big|_0^1 = \frac{1}{28}$.

   b. If the bounds on an integral include variables they have to be variables that haven't been integrated over yet.
      - Thus the final bounds must be constant.
   c. When choosing whether to integrate with respect to y first or with respect to x first, try to decide which will be easier.

II. Negative integrands
   a. What if the function $f(x,y)$ is positive over part of R and negative over the rest of R? (Remind students of single-variable case.)
   b. Then the value of $\iint_R f(x,y)\, dA$ is the volume under the graph of f and above the xy-plane minus the volume under the xy-plane and above the graph of f.
   c. Graphically:

III. More examples

a. Find the volume of the solid bounded above by $f(x, y) = 3xy$ and below by the region in the xy-plane lying between the graphs of $y = x$ and $y = x^2 - 4x + 4$.

i. Graphically:

ii. It will be easier to integrate this region with respect to y first.

1. If we integrate with respect to x first we would have to break the problem up into two double integrals.
2. This is because the x bounds are different depending on the particular value of y.
3. This region is what is known as *vertically simple*, as the same two curves mark the entering and leaving points of any vertical line through the region.

iii. The y bounds are $x^2 - 4x + 4$ and x.

iv. To find the x bounds we need to do some algebra.

1. Solve $x = x^2 - 4x + 4$.
2. This is $x^2 - 5x + 4 = 0 \implies (x - 4)(x - 1) = 0 \implies x = 1$ or $x = 4$.

v. Thus the integral is

$$\begin{aligned}
\int_1^4 \int_{x^2-4x+4}^{x} 3xy \, dy \, dx &= \frac{3}{2} \int_1^4 xy^2 \Big|_{x^2-4x+4}^{x} dx = \frac{3}{2} \int_1^4 x \left(x^2 - (x^2 - 4x + 4)^2\right) dx \\
&= \frac{3}{2} \int_1^4 x \left(x^2 - (x^4 - 8x^3 + 8x^2 + 16x^2 - 32x + 16)\right) dx \\
&= \frac{3}{2} \int_1^4 \left(-x^5 + 8x^4 - 23x^3 + 32x^2 - 16x\right) dx \\
&= \frac{3}{2} \left[-\frac{1}{6}x^6 + \frac{8}{5}x^5 - \frac{23}{4}x^4 + \frac{32}{3}x^3 - 8x^2\right]_1^4 \\
&= 60.075.
\end{aligned}$$

b. Evaluate the integral $\displaystyle\int_0^1 \int_x^1 \sin y^2 \, dy \, dx$.

i. This cannot be integrated as it stands.

ii. Perhaps if we switched the order of integration?

1. Graphically:

2. The region is both vertically and horizontally simple.
3. If we integrate first with respect to x and then with respect to y we have the integral $\displaystyle\int_0^1 \int_0^y \sin y^2 \, dx \, dy$.
4. This integral we can at least integrate with respect to x.

iii. We have

$$\int_0^1 \int_0^y \sin y^2 \, dx \, dy = \int_0^1 \left[x\right]_0^y \sin y^2 \, dy = \int_0^1 y \sin y^2 \, dy$$
$$= -\frac{1}{2} \cos y^2 \Big|_0^1 = \frac{1}{2}(1 - \cos 1).$$

iv. Sometimes switching the order of integration allows you to perform an integration you wouldn't be able to otherwise.

c. Find the volume of the region bounded by the cylinders $x^2+y^2 = r^2$ and $x^2+z^2 = r^2$.

i. By symmetry, the volume of the region is 8 times the volume contained solely in the first octant.

ii. But what does the region look like?

1. Discuss the projection of the solid onto each of the three coordinate planes.
2. The function that forms the top of the region is $z = \sqrt{r^2 - x^2}$.
3. The projection of the region onto the xy-plane looks like the following:

4. Show graph to students.

iii. We have $\iint_R \sqrt{r^2 - x^2} \, dA$.

iv. We should integrate with respect to y first.

v. Thus we have

$$\int_0^r \int_0^{\sqrt{r^2-x^2}} \sqrt{r^2 - x^2} \, dy \, dx = \int_0^r \sqrt{r^2 - x^2} \, y|_0^{\sqrt{r^2-x^2}} \, dx = \int_0^r (r^2 - x^2) \, dx$$
$$= r^2 x - \frac{1}{3}x^3 \Big|_0^r = r^3 - \frac{1}{3}r^3 - (0 - 0) = \frac{2}{3}r^3.$$

vi. Therefore, the entire volume is $\frac{16}{3}r^3$.

d. Find the volume of the region bounded above by the graph of $z = 2 - \sqrt{x^2 + y^2}$ and below by the graph of $z = x^2 + y^2$.

i. Graphically, the region is given by:

ii. This can be expressed as the volume of the region below the cone less the volume of the region below the paraboloid.

iii. But what region in the xy-plane are we integrating over?

1. We must find where the paraboloid and the cone intersect.
2. This will be in a circle of some sort.
3. Set the z-values equal to each other.
   - We have $2 - \sqrt{x^2 + y^2} = x^2 + y^2$.

- The circle that solves this equation is the circle $x^2 + y^2 = 1$.

4. Thus we're integrating over the following region:

5. Because of symmetry we can just integrate over the first quadrant and multiply by 4 to get the volume.

iv. Thus we're solving

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \left[2 - \sqrt{x^2 - y^2} - (x^2 + y^2)\right] dy\, dx.$$

v. This is really nasty – in fact, it's done more easily in polar coordinates, which we'll discuss soon.

IV. Density interpretation of an integral

a. What are the units of the integral $\iint_R f(x, y)\, dA$?

i. As in single-variable calculus, they are the units of f times the units of dA.

ii. For example, if f is in pounds per square feet, and dA is in square feet, then the integral is in pounds.

b. Thus, if f represents a density function – so much of a per so much of b – and the differential is in the same units as b, then the integral represents the total amount of a over the region R.

c. Examples (See Section 15.5, which we will skip)

i. In probability theory we discuss integrating probability density functions to obtain probabilities.

ii. You can integrate mass density functions to obtain total mass.

V. Average value

a. The average value of a single-variable function f over an interval $[a, b]$ is given by

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

b. Similarly, the average value of a function $f(x, y)$ over a region D is given by

$$\bar{f} = \frac{\iint_D f(x, y)\, dA}{\iint_D dA},$$

where $\iint_D dA$ is, again, the area of region D.

Homework: Section 15.2 #3, 6, 10, 19, 26, 33, 41, 45, 46, 52.